

Response Letter with Annotations

Duan, Z., Li, J., Lukito, J., Yang, K.-C., Chen, F., Shah, D. V., & Yang, S. (2022). Algorithmic agents in the hybrid media system: Social bots, selective amplification, and partisan news about COVID-19. *Human Communication Research*, 48(3), 516–542.
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Annotations

1. Purposes

a. Motivation (one-sentence summary)

To understand how social bots shape attention dynamics in the hybrid media system by selectively amplifying COVID-19 and influencing partisan news coverage.

b. Specific Rationale (several bullet points)

- Algorithmic bots are widespread and influential
- Most research focus on bot detection, but dynamic relationship among bots, news media, and the public remain unknown
- Development of natural language models

c. Research Questions and/or Hypotheses

• Identify IV, DV, mediator, moderator

This article has four research questions without proposing causal relationships, so there is no IV, DV, mediator or moderator.

• Identify specific relationships among variables

All four research questions are only descriptive.

RQ1 focuses on the information volume generated by bot-like accounts.
RQ2 focuses on topical foci of bot-like accounts across human accounts and media coverage.

RQ3 and RQ4 examine the bi-directional, over-time relationship between Twitter bots and human accounts/news coverage to identify which topics bots amplify and how that amplification subsequently shapes downstream human discourse/news coverage.

2. Method

a. Research Design

This research uses content analysis. It uses two datasets: (1) Twitter data obtained from Synthesio (a third-party provider) and (2) COVID-19 news stories. Both datasets cover the same three-month time span, from March 1, 2020, to May 31, 2020.

b. Sample (representative sample?)

For collecting their data, they develop and refined search keywords. Their sample is not representative, but appropriate to their research questions.

c. Measures

Bot detection:

- Botometer v4 with two thresholds (0.7-primary, 0.5-robustness check) to classify accounts as bots or human.
- Bots have higher cores.

Embedded topics across tweets and news stories: structural topic model

- They set $k = 50$

d. Analysis

Descriptive statistics for information volume (RQ1) and embedded topics (RQ2).
Time-series analysis to see the overtime relationship (RQ3, RQ4).

3. Results and Conclusions

a. Answers to RQs and Hypotheses

RQ1: bots make up 8.98% of total accounts in Covid-19 Twitter dataset. Despite a small number, bots are more active. Bots rely more on retweets than original posts or comments.

RQ2: Bots focus more on political/societal topics than human accounts. In addition, bots discuss more about three topics (Blaming China T3, Vaccine T30, and Senators Stock Scandal T41) than media outlets. Topical associations are less consistent when comparing Twitter bots with partisan media outlets, especially liberal media outlets

RQ3: Bots amplify topics have been discussed by humans. Human start to pay less attention to the topic that bots overwhelm the conversation.

RQ4: Bots do not amplify news coverage. Liberal and conservative outlets selectively increase coverage after bots amplify certain topics. Centrist outlets show little response to bots.

b. Include Key Figures/Tables

Table 1 Granger Causality Tests on the Topic 2 Panic Shopping

	X^2	p -value		X^2	p -value
Human $\rightarrow$ Bot	10.29	.001	Bot $\rightarrow$ Human	0.18	.673
Conservative media $\rightarrow$ Bot	0.01	.929	Bot $\rightarrow$ Conservative media	1.03	.311
Centrist media $\rightarrow$ Bot	0.85	.358	Bot $\rightarrow$ Centrist media	3.37	.067
Liberal media $\rightarrow$ Bot	2.10	.147	Bot $\rightarrow$ Liberal media	0.01	.911

Table 2 Summary of All Time-Series Analyses (with Bot Threshold at 0.7)

Human	Conservative	Central	Liberal	to Bots	Bots to	Human	Conservative	Central	Liberal
Daily Life (8)									
+10				Panic Shopping (T2)					+10
				Dog Immunity (T7)					+10
				Symptom Experience (T11)					+10
				Food & Cooking (T14)					+10
+10				School Closures & Changes (T19)					
+10				Critique of Wishful Thinking (T22)		-3			
+3				Hope & Faith (T26)					
+10		-10		National Sports Cancellation (T34)		-10			
Political/Societal (14)									
				Blaming China (T3)					
				Trump Response Failure (T4)					+10
+10		-4		Travel Bans & Global Lockdown (T6)		-10			
				Voting & Election (T16)					
				Hydroxychloroquine Misinformation (T17)					
		-10		Captain Crozier Scandal (T21)					+4
				Racism & Xenophobia (T25)			+10		
+4		-4		Economy & Market (T33)					
				Protest & Arrest (T36)					
				Senators Stock Scandal (T41)					
				Critiques of Media & Misinformation (T42)					
				Relief Bills (T43)					
+10				Call for Mitigation Strategies (T48)					
				Workers & Unemployment (T49)					

(Continued)

Table 2 (continued)

Human		Conservative	Central	Liberal	to Bots	Bots to	Human	Conservative	Central	Liberal			
Public Health (11)													
					Institutional Distrust & COVID Denial (T5)								
					Frontline Gratitude & Support (T12)								
					Case Reporting (T15)								
					Spring Break Super Spreader (T18)								
					Vaccine (T30)								
+3					Testing Availability (T31)				+2				
					Epicenter Cities (T38)								
					PPE & Hospital Supplies (T39)								
					Safety Orders & Guidelines (T45)								
+3					Masking & Personal Hygiene (T46)				+10				
					COVID-19 Research (T47)								

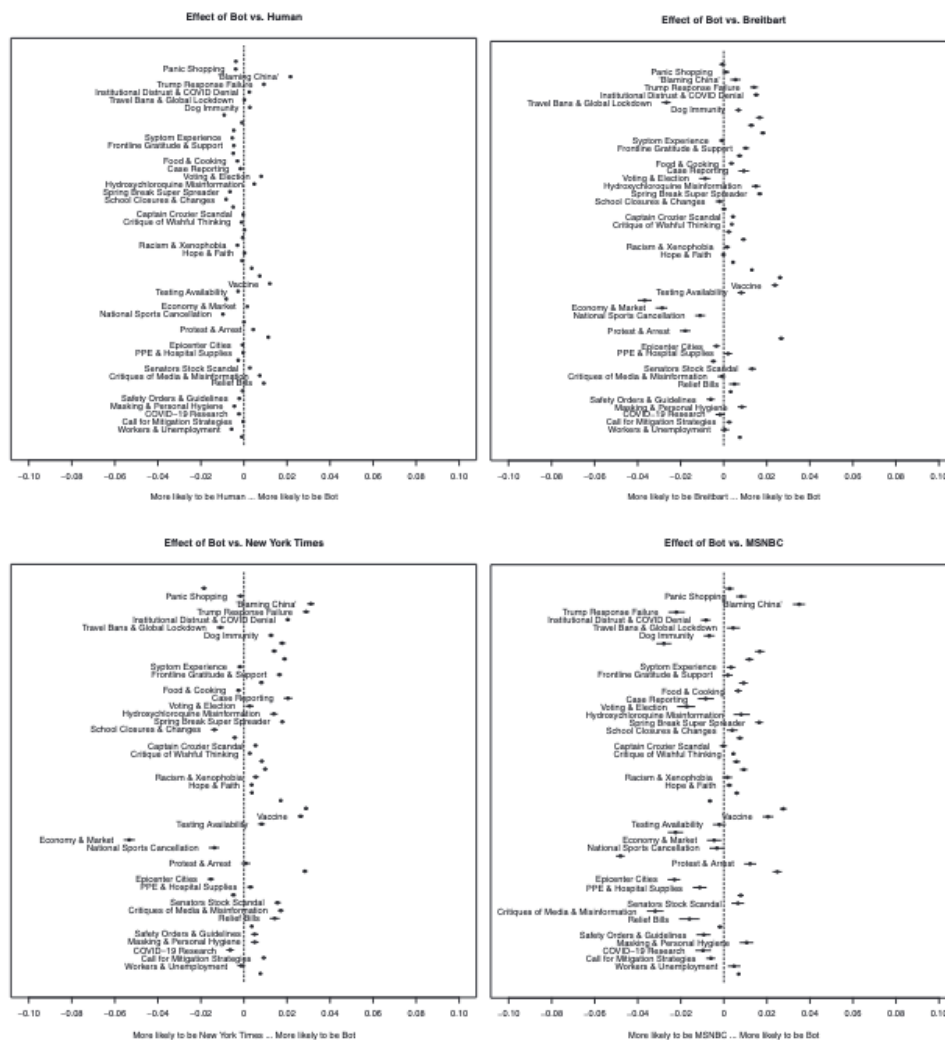


Figure 1 Topical Association Difference Between Twitter Bots (Right) and Some Non-Bot Actors (Left).

c. Authors' Conclusions

This research concludes that social bots don't function as a background noise but play an active and influential role in media environment. Even though bots make up a small number of accounts, they amplify selected human topics, especially

political, polarized, or high-attention issues. Also, social bots could influence partisan news coverage (liberal and conservative). However, human reduce their discourse after bots amplify on these topics.

4. Discussion

a. Implications

- Empirical evidence about how social bots works in media environment
- Extend the influence of social bots beyond Twitter
- Bot–human interactions reveal both amplification and avoidance

b. Limitations

- Bot detection is not perfect
- Focus on less complicated bots
- The design doesn't imply causality

c. Future Directions

- Look at other platforms or cross-platform dynamics
- Apply experiment for causality

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Response Letter (~600 words)

- **Your assessment of the study**

This research examines how social bots act as algorithmic agents within the hybrid media system in the early COVID-19 pandemic. The study finds that social bots, although a small share of accounts, are highly active amplifiers of human discourse, especially for politically, sensational, or high-attention topics. Social bots do not meaningfully influence human users, but they do shape the agendas of partisan news outlets. The authors conclude that bots are not background noise but algorithmic communicators that help transfer attention from social media into partisan news. Overall, the study is really rigorous and methodologically sophisticated.

This paper has several strengths. I think large-scale and multiple sources could enable cross-platform analysis. By using a large dataset of Twitter posts and news articles, the authors are able to compare the behavior of three different actors (bots, humans, and partisan news outlets) across 50 topics identified through structural topic modeling. I think previous literature often treat them separately. This comparison part is quite impressive. Furthermore, another strong feature is that the authors are very transparent about reporting bot detection and structural topic modeling in their appendix. Additionally, the longitudinal perspective introduced in RQ3 and RQ4 is innovative and makes it stand out. This aspect could tell how topics evolve overtime in these three actors and could get some sense of causality.

For weaknesses, authors mostly mention limitations in terms of their methodologies. For instance, Botometer may miss out more complicated bots, which means they might underestimate actual bot influence. Topic modeling procedure could introduce subjectivity, since it involves human interpretation and may introduce biases. Also, the temporal and topical scope is narrow in this research. The study only focuses on a relatively short time window (three-month period) and only focus on Covid-19, so their findings may not be generalizable to other time and context. Besides their limitations they acknowledge, I think there is a conceptual gap in the logic linking Twitter bots activity to news coverage, because bots and news operate on different platforms, the influence pathway is not straightforward. The study does not directly measure or validate whether newsrooms actually see or engage with bot-driven content. This makes the inferred connection weaker. Authors may think of some alternative ways, for example, analyze news organizations' official Twitter accounts, which could reveal whether journalists interact with or retweet bot-amplified content. They could also further validate these official Twitter accounts are consistent with news coverage under the same companies. Without this, I think the influence is not fully established.

For connection to related literature, this study builds on computational propaganda research by showing that bots do more than amplify political narratives on social media, but they also help shape news agendas, extending their influence beyond the platform. It also contributes to agenda-setting and intermedia agenda-setting scholarship. Their findings also resonate with other literature such as misinformation and polarization research, showing that partisan outlets selectively adopt narratives that reinforce existing identities. Finally, the study complicates prior work on social bots by adding human interactions.

Regarding their method, I think their multi-method pipeline allow this research to move beyond a simple text analysis. They start with bot detection, and then structural topic modelling to dividing topics, and finally use time-series analysis to examine temporal relationships among these three actors. However, while the time-series results suggest directional influence, they cannot establish causality. In the future, they could consider experimental or quasi-experimental designs to validate these patterns.